NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

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SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 9: L14

Question 1 (Lecture 14)

True or false: While the underrepresentation of specific groups in training data is a common contributor to data bias, addressing this imbalance alone is always sufficient to ensure equitable and unbiased performance of an AI model in real-world deployment scenarios.

(a) True

(b) False

Ans: (b)

Question 2 (Lecture 14)

Which of the following best describes model bias?

a) A biased dataset that leads to skewed predictions

b) A mismatch between training and deployment settings

c) A model that makes incorrect or overly complex assumptions, either oversimplifying or overfitting the data

d) A model trained using synthetic data

Ans: (c)

Question 3 (Lecture 14)

True or false: Model bias can occur even if the training data is perfectly balanced.

(a) True

(b) False

Ans: (a)

Question 4 (Lecture 14)

Which of the following practices most effectively mitigates model bias at its source?

a) Employing rigorous data standardization techniques to ensure all input features are scaled uniformly across the dataset

b) Maximizing the complexity of the chosen model architecture to allow for the learning of highly intricate and nuanced patterns within the training data

c) Consistently enriching the training dataset with new, contextually diverse, and representative examples, followed by iterative model retraining

d) Implementing post-hoc fairness algorithms to adjust the model's outputs only after initial predictions are generated, without altering the training process.

Ans: (c)

Question 5 (Lecture 14)

Which of the following is a key cause of deployment bias?

a) Using outdated training data

b) Selecting an overly simple model

c) Applying the model in a context very different from its training conditions

d) Ignoring missing values during data preprocessing

Ans: (c)

Question 6 (Optional – For Exploration)

An engineering firm is developing a new Large Language Model (LLM) application for use in confidential industrial design and simulation workflows. This LLM will process sensitive proprietary data and potentially intellectual property during its operation and refinement. The firm is acutely aware of the data privacy challenges inherent in LLMs.

Consider the following common privacy-enhancing techniques employed in LLM development and deployment:

- Data Anonymization
- Differential Privacy
- Access Controls and Filters
- Federated Learning / On-device processing

Select any two of these techniques. For each chosen technique, provide a detailed discussion of the key engineering **trade-offs** that must be evaluated during its design and implementation. Your analysis should encompass considerations such as:

--The impact on the LLM's performance (e.g., accuracy, model utility, inference speed).

--The user experience for end-users interacting with the LLM (e.g., ease of use, perceived privacy, functionality limitations).

--The system complexity introduced for development, integration, and maintenance.

--The guaranteed level of privacy assurance versus the feasibility and implementation cost.

Sample Answer:

Selected Techniques: Data anonymization & Federated Learning

1. Data Anonymization
Trade-offs:
Data anonymization removes or masks identifiable information before model training, protecting individual and organizational privacy. However, excessive anonymization can strip valuable context, reducing the LLM's accuracy and limiting its ability to learn domain-specific nuances. From a user's standpoint, anonymization strengthens confidence in data protection but may result in less precise or relevant responses. Engineering-wise, implementing robust anonymization pipelines adds preprocessing complexity and requires continuous updates to address re-identification risks as data and models evolve.

2. Federated Learning / On-device Processing
Trade-offs:
Federated learning allows model training or fine-tuning to occur locally on user devices or secure company servers, ensuring sensitive data never leaves its source. This approach significantly enhances privacy but increases system complexity, as it requires distributed coordination, version control, and secure communication protocols. Performance may also be affected by heterogeneous device capabilities or slower update cycles. For users, it offers high perceived privacy and ownership of data but may come with reduced responsiveness or higher latency in some workflows.

Takeaway

When deploying privacy-enhancing techniques for LLMs, engineers must carefully balance trade-offs. No free lunch in data privacy: improving privacy often impacts model performance, user experience, and system complexity. Selecting the right approach depends on the context, desired privacy guarantees, and practical feasibility.